# Single-Beacon Localization for Mobile Robot: A Set Membership Filtering Approach

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***Abstract*—In this letter, we study the localization problem of a mobile robot with range measurement from a single beacon. Previous filtering-based studies usually required accurate statistics of noises to be theoretically sound and reliable, which is difficult to obtain in practical systems. To solve this problem, we propose an accurate and efficient localization method based on a set-membership filtering framework with constrained zonotopes. This method includes three novel steps. First, we design a convex optimization relaxation method to handle the non-convexity caused by the single-beacon range measurement. Then, a halfspace-intersection refinement is proposed which improves the estimation accuracy. Finally, we provide a sliding-window recursive method that simultaneously guarantees the accuracy and the efficiency. Simulations and field experiments corroborate the effectiveness of our proposed method.**

***Index Terms*—Single-beacon localization, range measurement, set-membership filtering, constrained zonotope.**

## I. Introduction

RANGE-BASED localization for mobile robot systems has attracted significant research interest in recent years [1], [2], [3]. Different from the Global Positioning System (GPS)/inertial navigation [4] and the Vision-Based localization [5], range-based localization usually relies on wireless communications; thus, it can not only enable robotic localization in a wider range of environments but also improve localization accuracy through information fusion. Most of the current range-based localization methods through multi-beacon [6], require the installation of complex beacon networks in advance. However, if the beacon network can only be deployed temporarily while the robot is performing tasks, the network may only have one beacon available, which renders the existing methods inapplicable.

Single-beacon localization methods in robot systems have been studied recently in [7], [8], [9]. The main focus of the research is to fuse the single-beacon range measurement and the existing localization system to improve overall performance. For example, Yang et al. [7] improved the accuracy of the Inertial Measurement Unit (IMU) data via single-beacon ranging measurement; a sensor fusion method for Ultra-wideband (UWB) and Visual Odometry (VO) was presented in [8], which provided location information over unknown environments.

Note that sensor noise is a main factor affecting localization performance. In the literature, to attenuate the negative effects of the noise, the research of single-beacon localization relies on stochastic filters such as the Extended Kalman Filter (EKF) [10] or Unscented Kalman Filter (UKF) [11]. Stochastic filters perform well when the noise conforms to known distribution [12]. However, in real robot systems, the statistical characteristics of noise are assumed by approximation (e.g., Gaussian distribution); if this approximation significantly deviates from the real distribution, the filtering results are inaccurate [13].

Different from the stochastic filters, the Set-Membership Filterings (SMFs) [14], [15] only require the range of noises and without any statistical information, and the unknown-but-bounded (UBB) model is used to describe the noise. The SMF aims to estimate compact sets containing the true state of the mobile robot system. In recent years, many SMF state estimation methods with different set representation tools have been proposed. For example, Wang et al. [16] proposed an ellipsoidal SMF method with a new estimator, which improved the accuracy of estimation through effective parameters optimization. Tang et al. [17] provided an SMF method based on zonotopes, which minimized zonotopes via observer parameter optimization. Rego et al. [18] designed an SMF method based on constrained zonotopes, which recursively updated constrained zonotopes through mean value and first-order taylor extension.

Based on the theoretical breakthroughs, the SMF method has drawn increasing attention to solving the localization problem of mobile robots. A lane boundary augmented set-membership global navigation satellite systems positioning was presented in [19], which computed the absolute robotic position of geometric constraints by bounded noises of the sensor. A global set-membership filtering approach was proposed in [20], which obtained accurate localization results of automatic guided vehicles based on a distributed localization network. Nevertheless, research on the SMF localization method for the range measurement is currently limited. In [21], a constrain-convex-generator method was provided to deal with the non-convexity of the 2D range measurements by cutting the measurement annulus. Using a similar idea, [22] designed a cooperative SMF for distributed localization. Even though the above method can be applied to the single-beacon localization problem, unfortunately, it faces the issue of balancing the computational complexity and the accuracy. To the best of our knowledge, SMF localization problem for the single-beacon range measurement is still open.

In this letter, we propose an accurate and efficient method based on constrained zonotopes for the single-beacon localization problem with UBB noises. The main contributions are as follows:

Manuscript received 30 January 2024; accepted 6 May 2024. Date of publication 28 May 2024; date of current version 2 October 2024. This letter was recommended for publication by Associate Editor Y. Wang and Editor S. Behnke upon evaluation of the reviewers' comments. This work was supported by National Natural Science Foundation of China under Grant U23B2032 and Grant U2241214. *(Corresponding author: Jun Lai.)*

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This letter has supplementary downloadable material available at https://doi.org/10.1109/LRA.2024.3406190, provided by the authors.

Digital Object Identifier 10.1109/LRA.2024.3406190

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- To handle the non-convexity caused by the range measurement, we rigorously derive the mathematical description of the intersection determined by the range measurement and the location prior. Based on this description, an outer-bounding zonotope of the intersection is obtained by our convex relaxation method.
- With the zonotope, we design a halfspace-intersection refinement to improve the accuracy of the estimate. Simulation results show that our relaxation-refinement strategy outperforms the existing methods [21], [22].
- To further improve the efficiency of our algorithm while maintaining accuracy, a sliding window recursive method is proposed. Both the simulation and field experiments corroborate the effectiveness of our proposed method.

## II. System Model and Problem Formulation

Consider a mobile robot equipped with a range sensor (e.g. UWB [7]). The kinematic model and locational states of the mobile robot are described by the following equations of uncertain variables[1] ($k \in \mathbb{N}_0$):

$$\mathbf{x}_{k+1} = A_k \mathbf{x}_k + B_k \mathbf{w}_k, \tag{1}$$

$$\mathbf{z}_k = H \mathbf{x}_k, \tag{2}$$

which describe how the location $\mathbf{z}_k$ (with its realization $z_k \in [\![\mathbf{z}_k]\!] \subseteq \mathbb{R}^{n_z}$) changes over time, where $\mathbf{x}_k$ is the robot state (with its realization $x_k \in [\![\mathbf{x}_k]\!] \subseteq \mathbb{R}^{n_x}$); $\mathbf{w}_k$ is the process noise (with its realization $w_k \in [\![\mathbf{w}_k]\!] \subseteq \mathbb{R}^{n_w}$); $A_k \in \mathbb{R}^{n_x \times n_x}$, $B_k \in \mathbb{R}^{n_x \times n_w}$, and $H \in \mathbb{R}^{n_z \times n_x}$ are known matrices. Note that the linear model (1) can approximate nonlinear kinematics through linearization [24].

In this work, we focus on the single-beacon range measurement that can be modeled as:

$$\mathbf{y}_k = \|\mathbf{z}_k - p_k\| + \mathbf{v}_k, \quad k \in \mathbb{N}_0, \tag{3}$$

where the measurement $\mathbf{y}_k \in \mathbb{R}$ is a noise-corrupted distance between a known beacon $p_k \in \mathbb{R}^{n_z}$ (e.g. the location of a UWB beacon [7]) and the location of the robot $\mathbf{z}_k$; $\mathbf{v}_k$ is the measurement noise (with its realization $v_k \in [\![\mathbf{v}_k]\!] \subseteq \mathbb{R}^{n_v}$). The initial state $\mathbf{x}_0$, the process noises $\mathbf{w}_{0:k} := \mathbf{w}_0, \ldots, \mathbf{w}_k$, and the measurement noises $\mathbf{v}_{0:k} := \mathbf{v}_0, \ldots, \mathbf{v}_k$ are assumed to be unrelated.[2]

The set-membership localization problem is to derive an estimate $\mathcal{Z}_k$ that includes all possible locations $z_k$ given the measurements up to $k$, i.e., $y_{0:k}$. The optimal estimate is the smallest one which is exactly the conditional range $[\![\mathbf{z}_k|y_{0:k}]\!]$ [15]. An overview of the problem in Fig. 1.

When $\mathbf{x}_0$, $\mathbf{w}_{0:k}$, and $\mathbf{v}_{0:k}$ are unrelated, we can employ Theorem 2 in [15] to derive $[\![\mathbf{z}_k|y_{0:k}]\!]$; see Lemma 1.

*Lemma 1 (Set-Membership Localization Framework):* For $k \in \mathbb{N}_0$, the range of location $[\![\mathbf{z}_k|y_{0:k}]\!]$ can be recursively derived by the following steps:

- *Initialization:* Set the initial prior set $[\![\mathbf{x}_0]\!]$.
- *Prediction:* Given the posterior set $[\![\mathbf{x}_{k-1}|y_{0:k-1}]\!]$, the prior set $[\![\mathbf{x}_k|y_{0:k-1}]\!]$ is predicted by

$$[\![\mathbf{x}_k|y_{0:k-1}]\!] = A_{k-1}[\![\mathbf{x}_{k-1}|y_{0:k-1}]\!] \oplus B_{k-1}[\![\mathbf{w}_{k-1}]\!]. \tag{4}$$

[1]We use $[\![\mathbf{x}]\!]$ to represent the range of an uncertain variable [23] $\mathbf{x}$, i.e., $[\![\mathbf{x}]\!] := \{\mathbf{x}(\omega) : \omega \in \Omega\}$ and apply $x$ to denote the value in $\mathbf{x}$.

[2]This unrelatedness can be regarded as a non-stochastic version of the independence in stochastic model, defined in [15].

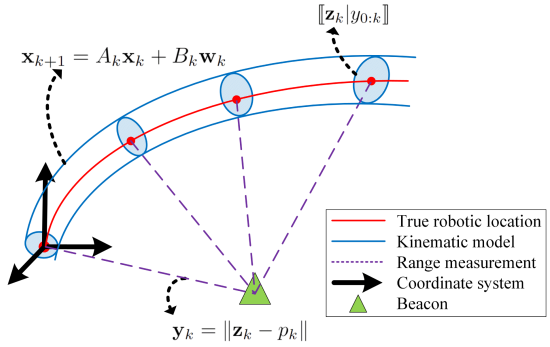



Fig. 1. Overview of the set-membership localization problem.

- *Update:* Given the measurement $y_k$ and the prior set $[\![\mathbf{x}_k|y_{0:k-1}]\!]$, the posterior set $[\![\mathbf{x}_k|y_{0:k}]\!]$ is updated by

$$[\![\mathbf{x}_k|y_{0:k}]\!] = [\![\mathbf{x}_k|y_{0:k-1}]\!] \bigcap\nolimits_H \mathcal{Y}(y_k, p_k, \mathbf{v}_k), \tag{5}$$

where $\mathcal{Y}(y_k, p_k, \mathbf{v}_k) = \{z_k \in \mathbb{R}^{n_z} : y_k = \|z_k - p_k\| + v_k,\ v_k \in [\![\mathbf{v}_k]\!]\}$, and $\bigcap_H$ denotes the generalized intersection [25].

Then, the location set is $[\![\mathbf{z}_k|y_{0:k}]\!] = H[\![\mathbf{x}_k|y_{0:k}]\!]$.

Because the exact description of $[\![\mathbf{z}_k|y_{0:k}]\!]$ is computationally intractable in general [26], we aim at computing an over-approximation of it (by $[\![\mathbf{z}_k|y_{0:k}]\!] \subseteq \mathcal{Z}_k$) in an accurate and efficient way. Since the non-convexity with the range measurement, the update step is key. Simultaneously, it is necessary to ensure the accuracy of localization estimation while the computational efficiency is guaranteed to solve online.

To achieve this objective, this letter designs a SMF-based localization method with constrained zonotopes. More details are shown in Section III.

## III. SMF-Based Localization Method With Constrained Zonotopes

In this article, we use (extended) constrained zonotopes in [27] to describe the outer bound calculated by Lemma 1, i.e., $\mathcal{Z}_k \supseteq [\![\mathbf{z}_k|y_{0:k}]\!]$ with the following form

$$\begin{aligned}\mathcal{Z}_k &:= Z\left(\hat{G}_k^z, \hat{c}_k^z, \hat{A}_k^z, \hat{b}_k^z, \hat{h}_k^z\right) \\ &= \left\{\hat{G}_k^z \xi + \hat{c}_k^z : \xi \in [\hat{h}_k^{z-}, \hat{h}_k^{z+}], \hat{A}_k^z \xi = \hat{b}_k^z\right\},\end{aligned}$$

where $(\hat{G}_k^z, \hat{c}_k^z, \hat{A}_k^z, \hat{b}_k^z) \in \mathbb{R}^{n_x \times n_g} \times \mathbb{R}^{n_x} \times \mathbb{R}^{n_c \times n_g} \times \mathbb{R}^{n_c}$, and $\hat{h}_k^z \in [\hat{h}_k^{z-}, \hat{h}_k^{z+}] := \{\xi \in \mathbb{R}^{n_g} : \hat{h}_k^{z-,i} \le \xi_i \le \hat{h}_k^{z+,i}\}$ refers to the interval vector of dimension $n_g$; $n_g$ and $n_c$ are the generator and constraint dimensions [25], respectively. Constrained zonotope operations are defined in [25] and [27]. When $\hat{A}_k^z$ and $\hat{b}_k^z$ are empty and $\hat{h}_k^z$ is a unit interval constraint, constrained zonotope degenerates to zonotope and is denoted by the symbol $Z(\hat{R}_k^z, \hat{S}_k^z)$. Similarly, $[\![\mathbf{x}_k|y_{0:k-1}]\!] \subseteq \bar{\mathcal{X}}_k := Z(\bar{G}_k^x, \bar{c}_k^x, \bar{A}_k^x, \bar{b}_k^x, \bar{h}_k^x)$ and $[\![\mathbf{x}_k|y_{0:k}]\!] \subseteq \hat{\mathcal{X}}_k := Z(\hat{G}_k^x, \hat{c}_k^x, \hat{A}_k^x, \hat{b}_k^x, \hat{h}_k^x)$ and $[\![\mathbf{w}_k]\!] \subseteq \mathcal{W}_k := Z(\hat{G}_k^w, \hat{c}_k^w, \hat{A}_k^w, \hat{b}_k^w, \hat{h}_k^w)$.

Corresponding to Lemma 1, Fig. 2 shows the localization process of our method, which includes four parts as follows.

*1) Prediction:* Given the posterior set $\hat{\mathcal{X}}_{k-1}$ obtained at the previous time step $k-1$, the prediction set $\bar{\mathcal{X}}_k = A_{k-1}\hat{\mathcal{X}}_{k-1} \oplus$

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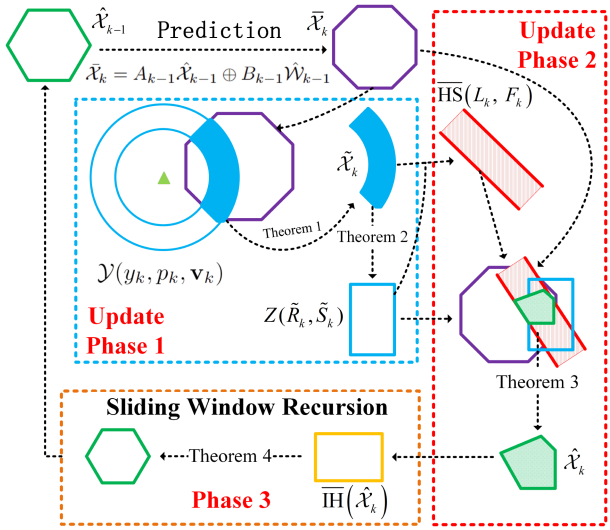



Fig. 2. Process of SMF-based localization method. The green hexagon denotes the posterior set $\hat{\mathcal{X}}_{k-1}$; the purple octagon denotes the prior set $\bar{\mathcal{X}}_k$; the blue solid rectangle denotes the $Z(\tilde{R}_k, \tilde{S}_k)$; the green shaded pentagon denotes the posterior set $\hat{\mathcal{X}}_k$. The $\overline{\text{IH}}(\hat{\mathcal{X}}_k)$ denotes the interval hull of $\hat{\mathcal{X}}_k$, which is obtained by linear programming (LP) [25]. The three dashed boxes represent Section III-A, Section III-B and Section III-C, respectively.

$B_{k-1}\hat{\mathcal{W}}_{k-1}$ is constructed to outer bound $[\![\mathbf{x}_k | y_{0:k-1}]\!]$ at time step $k$. *2) Update phase 1:* Given $\mathcal{Y}(y_k, p_k, \mathbf{v}_k)$ determined by a single-beacon range measurement and the prediction set $\bar{\mathcal{X}}_k$, an outer-bounding zonotope of $\tilde{\mathcal{X}}_k := \bar{\mathcal{X}}_k \bigcap_H \mathcal{Y}(y_k, p_k, \mathbf{v}_k)$, i.e., $\tilde{\mathcal{X}}_k \subseteq Z(\tilde{R}_k, \tilde{S}_k)$, is calculated. *3) Update phase 2:* Given $\bar{\mathcal{X}}_k$, $Z(\tilde{R}_k, \tilde{S}_k)$, $\mathcal{Y}(y_k, p_k, \mathbf{v}_k)$, the posterior set $\hat{X}_k$ is computed by constrained zonotope intersection. *4) Sliding-window recursion:* A sliding window recursive method is proposed to guarantee the efficiency and accuracy of the overall recursion.

### A. Phase 1: Convex Relaxation Approximation

In the convex relaxation approximation phase (the blue dotted squares in Fig. 2), given $\bar{\mathcal{X}}_{k-1}$, $\mathcal{Y}(y_k, p_k, \mathbf{v}_k)$, a convex relaxation approximation is proposed to obtain the outer-bounding zonotope of $\tilde{\mathcal{X}}_k$ (with $\bar{\mathcal{X}}_k \bigcap_H \mathcal{Y}(y_k, p_k, \mathbf{v}_k) = \tilde{\mathcal{X}}_k \subseteq Z(\tilde{R}_k, \tilde{S}_k)$). The update set $\tilde{\mathcal{X}}_k$ is formed by the intersection of the prediction set $\bar{\mathcal{X}}_k$ and $\mathcal{Y}(y_k, p_k, \mathbf{v}_k)$. It is a non-convexity set, the specific description and explanation about it is shown in Theorem 1.

*Theorem 1:* Given $\bar{\mathcal{X}}_k = Z(\bar{G}_k^x, \bar{c}_k^x, \bar{A}_k^x, \bar{b}_k^x, \bar{h}_k^x)$ and $\mathcal{Y}(y_k, p_k, \mathbf{v}_k)$, $\tilde{\mathcal{X}}_k = \bar{\mathcal{X}}_k \bigcap_H \mathcal{Y}(y_k, p_k, \mathbf{v}_k)$ is non-convex set,

$$\tilde{\mathcal{X}}_k = \left\{ H\bar{G}_k^x \xi + H\bar{c}_k^x : \xi \in [\bar{h}_k^{x-}, \bar{h}_k^{x+}], \bar{A}_k^x \xi = \bar{b}_k^x, \right.$$
$$\left. \xi^T Q_k^- \xi + q_k^- \xi + d_k^- \leqslant 0, \xi^T Q_k^+ \xi + q_k^+ \xi + d_k^+ \leqslant 0 \right\}, \tag{6}$$

$$Q_k^- = -\sum_{i=1}^{n_z} \left( (\bar{g}_k^i)^T \bar{g}_k^i \right), \; Q_k^+ = \sum_{i=1}^{n_z} \left( (\bar{g}_k^i)^T \bar{g}_k^i \right),$$

$$q_k^- = -2\sum_{i=1}^{n_z} (\bar{c}_k^i - p_k^i)\bar{g}_k^i, \; q_k^+ = 2\sum_{i=1}^{n_z} (\bar{c}_k^i - p_k^i)\bar{g}_k^i,$$

$$d_k^- = (y_k^-)^2 - \sum_{i=1}^{n_z} \left(\bar{c}_k^i - p_k^i\right)^2, \; d_k^+ = \sum_{i=1}^{n_z} \left(\bar{c}_k^i - p_k^i\right)^2 - (y_k^+)^2,$$

$$y_k^- = y_k - \inf_{v_k \in [\![\mathbf{v}_k]\!]} v_k, \; y_k^+ = y_k - \sup_{v_k \in [\![\mathbf{v}_k]\!]} v_k,$$

$\bar{g}_k^i$ is the $i$-th row of $H\bar{G}_k^x$, $\bar{c}_k^i$ is the $i$-th element of $H\bar{c}_k^x$, and $p_k^i$ is the $i$-th element of $p_k$.

*Proof:* See Appendix A. ■

Theorem 1 provides the specific form of $\tilde{\mathcal{X}}_k$ and shows that $\tilde{\mathcal{X}}_k$ is a non-convex set. This implies that this subsequent representation cannot be described via the constrained zonotope [25], and non-convexity leads to unnecessary computational complexity. Therefore, we need to provide a convex relaxation representation of $\tilde{\mathcal{X}}_k$. In this letter, we propose a convex optimization to compute $\tilde{\mathcal{X}}_k \subseteq Z(\tilde{R}_k, \tilde{S}_k)$, which is specifically computed as in Theorem 2.

*Theorem 2:* $\tilde{\mathcal{X}}_k = \bar{\mathcal{X}}_k \bigcap_H \mathcal{Y}(y_k, p_k, \mathbf{v}_k)$ is outer bounded by $Z(\tilde{R}_k, \tilde{S}_k)$ with

$$\tilde{R}_k = \begin{bmatrix} \frac{\rho_k^{1+} - \rho_k^{1-}}{2} & & \\ & \ddots & \\ & & \frac{\rho_k^{n_z+} - \rho_k^{n_z-}}{2} \end{bmatrix}, \tilde{S}_k = \begin{bmatrix} \frac{\rho_k^{1-} + \rho_k^{1+}}{2} \\ \vdots \\ \frac{\rho_k^{n_z-} + \rho_k^{n_z+}}{2} \end{bmatrix}, \tag{7}$$

where $\rho_k^{i-}$, $\rho_k^{i+}$ denote the minimum and maximum values of $\tilde{X}_k$ in dimension $i$, respectively. $\rho_k^{i-}$ is obtained by the following convex optimization [$\rho_k^{i+}$ is obtained by maximizing the objective (8)]

$$\min \; \rho_k^{i-} = \bar{g}_k^i \xi + \bar{c}_k^i \tag{8}$$

$$\text{s.t. } \xi^T Q_k^* \xi + q_k^* \xi + d_k^* \leqslant 0, \tag{9}$$

$$\xi^T Q_k^+ \xi + q_k^+ \xi + d_k^+ \leqslant 0, \tag{10}$$

$$\xi \in \left[\bar{h}_k^{x-}, \bar{h}_k^{x+}\right], \tag{11}$$

$$\bar{A}_k^x \xi = \bar{b}_k^x, \tag{12}$$

where

$$Q_k^* = \sum_{\lambda_i \geqslant 0}^{n_z} \lambda_i \nu_i \nu_i^T, \tag{13}$$

$$q_k^* = \sum_{\lambda_i < 0}^{n_z} \lambda_i \left(a_i^- + a_i^+\right) \nu_i^T + q_k^-, \tag{14}$$

$$d_k^* = d_k^- - \sum_{\lambda_i < 0}^{n_z} \lambda_i a_i^- a_i^+, \tag{15}$$

$$a_i^- = \nu_i^T \bar{h}_k^{x-}, \; a_i^+ = \nu_i^T \bar{h}_k^{x+}, \tag{16}$$

$$Q_k^- = \left[\nu_1, \ldots \quad \nu_{n_z}\right] \text{diag}\left[\lambda_1, \ldots, \lambda_{n_z}\right] \left[\nu_1, \ldots, \nu_{n_z}\right]^T. \tag{17}$$

Equation (17) denotes the eigenvalue decomposition of $Q_k^-$. In the above optimization process, $Q_k^*$ and $Q_k^+$ both are positive semi-definiteness condition.

*Proof:* See Appendix B. ■

*Remark 1:* Theorem 2 provides a convex optimization method for solving the outer bound of $\tilde{\mathcal{X}}_k$. The constraint (9), (10) in Theorem 2 are transformed into quadratic cone constraints, i.e., $\xi^T Q_k^* \xi + q_k^* \xi + d_k^* \leqslant 0$ is equivalent to $\sqrt{\|(Q_k^*)^{\frac{1}{2}} \xi\|^2 + (\frac{1 + \frac{1}{2} d_k^* + \frac{1}{2}(q_k^*)^T \xi}{\sqrt{2}})^2} \leq \frac{1 - \frac{1}{2} d_k^* - \frac{1}{2}(q_k^*)^T \xi}{\sqrt{2}}$, here

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$(Q_k^*)^{\frac{1}{2}}$ is a real symmetric square matrix such that $Q_k^* = (Q_k^*)^{\frac{1}{2}}(Q_k^*)^{\frac{1}{2}}$. The quadratic cone constraint can be solved quickly by the interior point method, and this ensures the overall computational efficiency of set-membership localization.

### B. Phase 2: Refinement of Intersection

In the refinement of intersection phase (the red dotted squares in Fig. 2), given $\bar{\mathcal{X}}_k$, $Z(\tilde{R}_k, \tilde{S}_k)$, and $\mathcal{Y}(y_k, p_k, \mathbf{v}_k)$, the goal is to construct the posterior set $\hat{\mathcal{X}}_k$ via the constrained zonotope intersection.

We define a halfspace constraint[3] $\overline{\text{HS}}(L_k, F_k)$ to refine the relaxed part of $Z(\tilde{R}_k, \tilde{S}_k)$ to ensure localization accuracy, and $\overline{\text{HS}}(L_k, F_k)$ is obtained by the following steps. Intersect $Z(\tilde{R}_k, \tilde{S}_k)$ and the equational constraints of $\mathcal{Y}(y_k, p_k, \mathbf{v}_k)$ to obtain the intersection points ${\mathcal{V}^-}_k$ and ${\mathcal{V}^+}_k$.

$$\mathcal{V}_k^- = \left\{ \tilde{R}_k\xi + \tilde{S}_k : \xi^T Q_k^- \xi + q_k^- \xi + d_k^- = 0, \|\xi\|_\infty \leqslant 1 \right\},$$

$$\mathcal{V}_k^+ = \left\{ \tilde{R}_k\xi + \tilde{S}_k : \xi^T Q_k^+ \xi + q_k^+ \xi + d_k^+ = 0, \|\xi\|_\infty \leqslant 1 \right\}.$$

The halfspace constraint $\overline{\text{HS}}(L_k, F_k)$ is computed through the intersection points, see [29] for details. Eventually, the posterior set $\hat{\mathcal{X}}_k$ is obtained by constrained zonotope intersection [25] and halfspace intersection [28], $\hat{\mathcal{X}}_k = \bar{\mathcal{X}}_k \bigcap_H (\overline{\text{HS}}(L_k, F_k) \bigcap Z(\tilde{R}_k, \tilde{S}_k))$, and Theorem 3 expresses the concrete intersection process.

*Theorem 3:* Given $\bar{\mathcal{X}}_k = Z(\bar{G}_k^x, \bar{c}_k^x, \bar{A}_k^x, \bar{b}_k^x, \bar{h}_k^x)$, $Z(\tilde{R}_k, \tilde{S}_k)$, and $\overline{\text{HS}}(L_k, F_k)$, the posterior set $\hat{\mathcal{X}}_k$ is obtained in the following form.

$$\begin{aligned}
\hat{\mathcal{X}}_k &= Z\left(\hat{G}_k, \hat{c}_k, \hat{A}_k, \hat{b}_k, \hat{h}_k\right) \\
&= Z\left( \begin{bmatrix} \bar{G}_k^x & 0 \end{bmatrix}, \bar{c}_k^x, \begin{bmatrix} \bar{A}_k^x & 0 \\ 0 & \begin{bmatrix} L_k\tilde{R}_k & D_k \end{bmatrix} \\ H\bar{G}_k^x & -\begin{bmatrix} \tilde{R}_k & \mathbf{0} \end{bmatrix} \end{bmatrix}, \right. \\
&\quad \times \begin{bmatrix} & \bar{b}_k^x & \\ F_k - L_k\tilde{S}_k - \frac{1}{2}\begin{bmatrix} d_k^1 & \dots & d_k^{n_p} \end{bmatrix}^T \tilde{S}_k - H\bar{c}_k^x & & \end{bmatrix}, \\
&\quad \left. \times \begin{bmatrix} \bar{h}_k^x \\ I_{(n_z+n_p)\times 1} \end{bmatrix} \right), \\
D_k &= \frac{1}{2}\text{diag}\left(\begin{bmatrix} d_k^1 & \dots & d_k^{n_p} \end{bmatrix}\right), d_k^i = f_k^i - l_k^i\tilde{S}_k + \sum_{j=1}^{n_z} \left| l_k^i r_k^j \right|,
\end{aligned} \tag{18}$$

where $n_p$ denotes the number of halfspaces, $l_k^i$ and $f_k^i$ are parameters of the $i$-th planar constraints in $\overline{\text{HS}}(L_k, F_k)$, and $r_k^j$ is the $j$-th row of $\tilde{R}_k$.

*Proof:* See Appendix C. ■

*Remark 2:* Theorem 3 describes how to compute the estimate $\hat{X}_k$ of $[\![\mathbf{x}_k | y_{0:k}]\!]$ in (5). An intuitive presentation in a 2D plane is shown in Fig. 3, without loss of generality, we assume beacon is the origin (with $p_k = \begin{bmatrix} 0 & 0 \end{bmatrix}^T$).

[3] $\overline{\text{HS}}(L_k, F_k) = \{\rho \in \mathbb{R}^{n_z} : l_k^i \rho \leq f_k^i, l_k^i \in L_k, f_k^i \in F_k, i = 1, \dots, n_p\}$ is a tool for describing uncertain range [28].

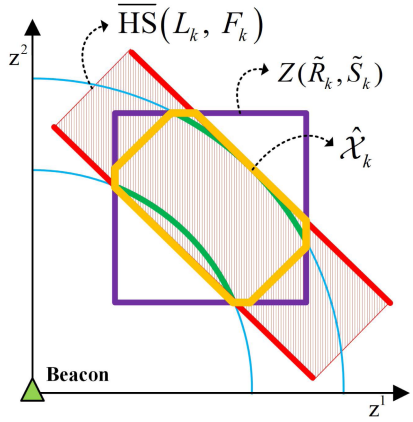



Fig. 3. A 2D example of Theorem 3. The green part denotes $[\![\mathbf{x}_k | y_{0:k}]\!]$; the red shaded part represents halfspace $\overline{\text{HS}}(L_k, F_k)$; the purple square represents the zonotope $Z(\tilde{R}_k, \tilde{S}_k)$; the yellow polygon represents the posterior set $\hat{\mathcal{X}}_k$. $z^1$ and $z^2$ are the coordinates.

*Remark 3:* The segmentation-based approach method [21] and [22] divides the range measurement annulus into $n$ chunks to obtain the posterior set $\hat{\mathcal{X}}_k$ in 2D space. The estimation accuracy and the computational efficiency are proportional to $1/n$ and $n$, respectively. This means increasing the accuracy must sacrifice the efficiency, and vice versa. In contrast, our method can simultaneously have high accuracy and efficiency (see Fig. 5 in Section IV-A1). Moreover, different from the method in [21] and [22], our proposed method can handle the localization problem in 3D space.

### C. Phase 3: Sliding Window Recursive Method

In the sliding window recursive method phase (the yellow dotted squares in Fig. 2), a fast and efficient recursive method is designed to ensure computational efficiency. To start with, we provide two definitions as follows.

*Definition 1 (SMF Map):* At time $k \in \mathbb{N}_0$, the SMF map is

$$\Gamma_k : \hat{\mathcal{X}}_{k-1} \mapsto \hat{\mathcal{X}}_k, \tag{19}$$

where $\Gamma_k$ denotes a one-step complete recursion containing prediction (4) and update (5), such that $\hat{\mathcal{X}}_{k-1}$ maps to $\hat{\mathcal{X}}_k$.

*Definition 2 (Sliding Window Map):* The sliding window map $\mathcal{F}_{k-\delta:k}$ with the sliding window side $\delta \leq k$ is

$$\mathcal{F}_{k-\delta:k} = \Gamma_{k-\delta+1} \circ \Gamma_{k-\delta+2} \cdots \circ \Gamma_k, \tag{20}$$

where $\circ$ is the function composition.

With the help of Definition 1 and Definition 2, we obtain Theorem 4.

*Theorem 4:* For the sliding window size $\delta \leq k$,

$$\hat{\mathcal{X}}_k = Z\left(\hat{G}_k^x, \hat{c}_k^x, \hat{A}_k^x, \hat{b}_k^x, \hat{h}_k^x\right) \subseteq \mathcal{F}_{k-\delta:k}\left(\overline{\text{IH}}\left(\hat{\mathcal{X}}_{k-\delta}\right)\right), \tag{21}$$

where $\overline{\text{IH}}(\hat{\mathcal{X}}_{k-\delta})$ is the interval hull of $\hat{\mathcal{X}}_{k-\delta}$.

*Proof:* See Appendix D. ■

According to Theorem 4, we design the Algorithm 1.

*Remark 4:* Existing studies commonly used reduction methods (e.g., in [25]) to improve the computational efficiency.

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**Algorithm 1:** Sliding Window Recursive Algorithm.

**Initialization:** Bounded $[\![\mathbf{x}_0]\!] = Z(\hat{G}_0^x, \hat{c}_0^x, \hat{A}_0^x, \hat{b}_0^x, \hat{h}_0^x)$, select sliding window size $\delta$.
**loop**
  $\overline{\mathrm{IH}}(\hat{\mathcal{X}}_{k-1}) \leftarrow \hat{\mathcal{X}}_{k-1}$ by linear programming in [25].
  **if** $k \leqslant \delta$ **then**
    $Z(\hat{G}_k^x, \hat{c}_k^x, \hat{A}_k^x, \hat{b}_k^x, \hat{h}_k^x) \leftarrow \Gamma_k(\hat{\mathcal{X}}_{k-1})$ by Lemma 1.
  **else if** $k > \delta$ **then**
    $Z(\hat{G}_k^x, \hat{c}_k^x, \hat{A}_k^x, \hat{b}_k^x, \hat{h}_k^x) \leftarrow \mathcal{F}_{k-\delta:k}(\overline{\mathrm{IH}}(\hat{\mathcal{X}}_{k-\delta}))$ by Section III.
  **end if**
  $\hat{\mathcal{X}}_k = Z(\hat{G}_k^x, \hat{c}_k^x, \hat{A}_k^x, \hat{b}_k^x, \hat{h}_k^x)$.
  $\hat{\mathcal{Z}}_k = HZ(\hat{G}_k^x, \hat{c}_k^x, \hat{A}_k^x, \hat{b}_k^x, \hat{h}_k^x)$.
  $k++$
**end loop**

When posterior set $\hat{\mathcal{X}}_k = Z(\hat{G}_k, \hat{c}_k, \hat{A}_k, \hat{b}_k, \hat{h}_k) \subset \mathbb{R}^{n_x}$ and $\hat{A}_k \in \mathbb{R}^{n_c \times n_g}$, the worst-case computational complexity for dimensionality reduction is $O(k_c(n_c+n_g)^3)$. The sliding window computational complexity is $O(\delta n_x^3)$. Because $\delta$ and $n_x$ are much smaller than $k_c$, $n_h$ and $n_g$, the sliding window recursive method has higher efficiency.

## IV. Simulation and Field Experiment

### A. Numerical Examples: Ablation Experiments

Two 2-D numerical ablation experiments are used to demonstrate the high accuracy and efficiency of the proposed method. The common parameter settings for both experiments are:

$$A_k = \begin{bmatrix} 0.6 - 0.1\sin(0.1\ k) & 0 \\ -0.2\sin(0.2\ k) & 0.4 \end{bmatrix},$$

$$B_k = \begin{bmatrix} 1 & 0 \\ 0 & \sin(0.15\ k) \end{bmatrix},\ H = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$$

$$[\![\mathbf{v}_k]\!] = [-0.2, 0.2],$$

$$[\![\mathbf{w}_k]\!] = Z\left(\begin{bmatrix} 5\sin(0.15\ k) & 5 \end{bmatrix}^T,\right.$$
$$\left.\mathrm{diag}\left(\begin{bmatrix} 7+7\sin(k) & 7\csc(0.15\ k)+7\cos(k)\csc(0.15\ k) \end{bmatrix}\right)\right). \quad (22)$$

The initial state is $x_0 = \begin{bmatrix} 0 & 0 \end{bmatrix}^T$; we set $[\![\mathbf{x}_0]\!] = Z(\begin{bmatrix} -0.1 & 0.1 \end{bmatrix}^T, \mathrm{diag}(\begin{bmatrix} 1 & 1 \end{bmatrix}))$ and $p = \begin{bmatrix} 0 & -50 \end{bmatrix}^T$.[4]

*1) Non-Convexity Measurement:* In this simulation, Algorithm 1 was compared with the EKF method and other SMF methods (the constrained convex generators (CCG) in [21], [22]), where the point estimation for SMF methods are the centroid of the interval hull of the set estimate. For the CCG method, the number of chunks $n$ is set to 25, and the sliding window size in Algorithm 1 is $\delta = 3$. The simulation results are shown in Fig. 4 and Table I, where MAE is the mean absolute error. The estimation results obtained by Algorithm 1 are more accurate and faster than those by the CCG method. Also, Algorithm 1 outperforms the EKF with the best parameter after tuning.

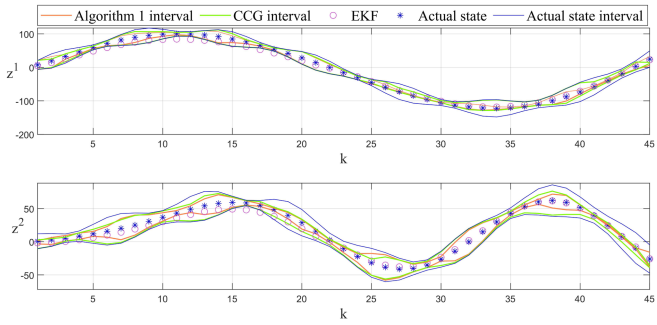



Fig. 4. The estimation results for the experiment in Section IV-A1. Besides point estimates, the SMF methods can also provide set estimates.

TABLE I
Comparison for the Experiment in Section IV-A1

| Method | $z^1$ range | $z^2$ range | MAE | average time |
|---|---|---|---|---|
| CCG | 21.7 | 19.3 | 4.1 | 23.9ms |
| Algorithm 1 | **14.6** | **13.9** | **1.7** | **2.6**ms |
| EKF | - | - | 4.71 | - |

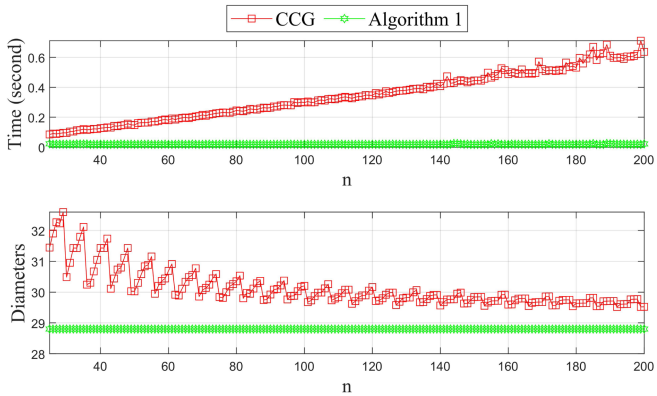



Fig. 5. Comparision between the CCG method and Algorithm 1 at $k = 20$.

Furthermore, we compare Algorithm 1 with the CCG method for a fixed time instant ($k = 20$). For the CCG method, the number of chunking $n$ varies from 25 to 200. The computational time and diameters[5] of the posterior are depicted in Fig. 5. From Fig. 5, we can see that the estimation accuracy and computational time of the CCG method increase with $n$. When $n = 25$, the estimated result diameter is 31.7, and the time required is 65.877 ms; when $n = 200$, the estimated result diameter is 29.5, and the time required is 636.9 ms. Compared to the CCG method, our Algorithm 1 has higher accuracy (with diameter 28.8) and efficiency (1.94 ms).

*2) Sliding Window Recursive Method Versus Existing Reduction Methods:* Using the same relaxation-refinement method in Section III-A and Section III-B, we compare our sliding window method in Algorithm 1 with the existing reduction methods.

[4]Note that the uncertain variables $\mathbf{x}_0$, $\mathbf{w}_{0:k}$, and $\mathbf{v}_{0:k}$ are uniformly distributed in their ranges.

[5]The diameter is in the sense of $\infty$-norm, and larger diameters indicate lower estimation accuracy.

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TABLE II
COMPARISON FOR THE EXPERIMENT IN SECTION IV-A2

| Method | Parameters | $z^1$ | $z^2$ | time |
|---|---|---|---|---|
| Reduction (girard) | (3, 5) | 5.57 | 4.99 | 52.6ms |
| Reduction (girard) | (5, 10) | 5.17 | 4.47 | 141.4ms |
| Reduction (combastel) | (3, 5) | 5.65 | 5.39 | 56.5ms |
| Reduction (combastel) | (5, 10) | 5.26 | 5.39 | 146.4ms |
| Algorithm 1 | $\boldsymbol{\delta = 3}$ | **4.59** | **4.37** | $\boldsymbol{1.79ms}$ |
| Algorithm 1 | $\delta = 4$ | 4.59 | 4.37 | 3.90ms |

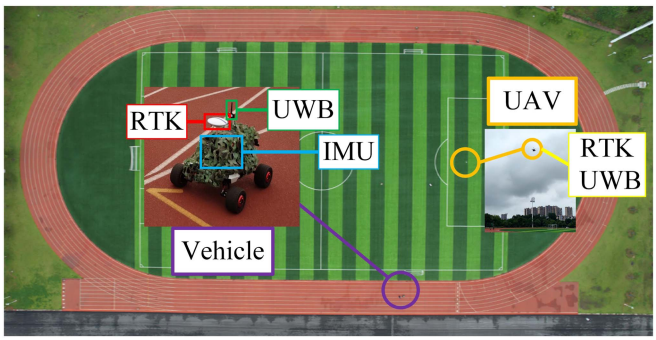



Fig. 6. The scenario and equipment for the air-to-ground localization. The UAV is equipped with a UWB receiver and a real-time kinematic (RTK) receiver, where the RTK provides the precise location for the UAV. The UGV is equipped with an IMU, a UWB receiver, and a GPS antenna. The GPS/IMU integrated navigation provided the unmanned vehicle status prediction, and UWB measurement from the UAV to the unmanned vehicle.

TABLE III
COMPARISON BETWEEN ALGORITHM 1 AND EKFS FOR THE AIR-TO-GROUND LOCALIZATION

| Method | Algorithm 1 | r=0.01 | r=0.001 | r=0.0005 |
|---|---|---|---|---|
| MAE (m) | 3.42 | 5.17 | 3.30 | 5.81 |

The results are shown in Table II: the first two columns provide the algorithm information (i.e., the "girard" and "combastel" methods in the toolbox CORA 2020 [30] and Algorithm 1, with different parameters, where $o_c$ and $o_d$ are the constraint dimension and degrees-of-freedom order [25], respectively); columns 3-4 give the lengths of the estimate w.r.t. $z^1$ and $z^2$ axes, respectively; the averaged computation time of the constrained zonotopes in each time step is shown in the last column. We can see that our algorithm with $\delta = 3$ outperforms the existing methods, and there is no need to further enlarge $\delta$.

### B. Air-to-Ground Localization

The experiment of locating an unmanned ground vehicle (UGV) by unmanned aerial vehicle (UAV) was conducted in the field, as shown in Fig. 6. We compare our proposed Algorithm 1 with the EKF with different parameters $r$ (the measurement error matrix in the EKF, which is a scalar in this case). The precise and estimated trajectories are shown in Fig. 7; the comparison w.r.t. the MAE is given in Table III.

As shown in Table III, the performance of the EKF depends on the statistical information on the measurement noises; if $r$ cannot reflect the truth, the MAE decreases; in contrast, Algorithm 1 does not rely on this statistical information, and its performance is very close to the EKF with the optimal parameter $r = 0.001$.

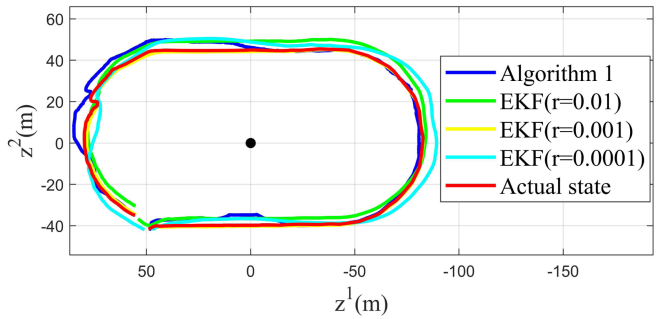
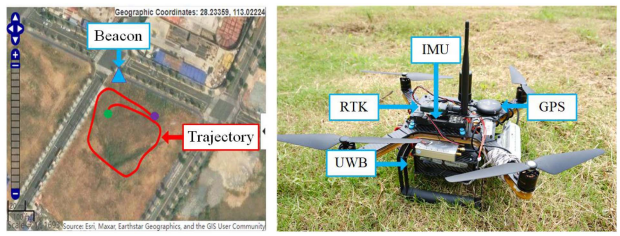
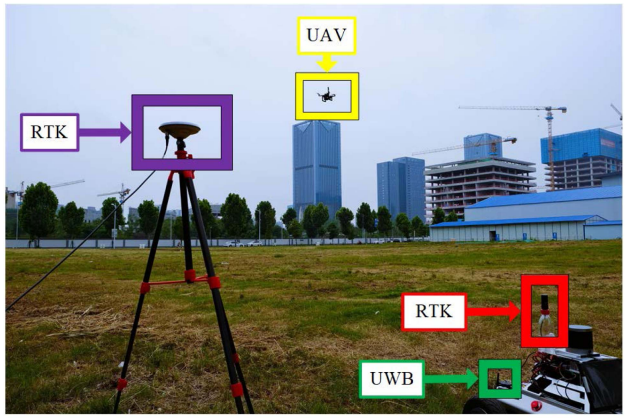



Fig. 7. Trajectories of the UGV for the air-to-ground localization.




(a) UAV trajectory. (b) UAV and sensors.




(c) Field experiment scenario.

Fig. 8. Experimental UAV and trajectory. In (a), the purple point is the starting point, and the green point is the ending point.

### C. Field Flight Experiment: UAV Localization

To further verify the effectiveness of Algorithm 1, a field experiment was carried out. A UAV equipped with low-precision IMU and GPS moves within the range that can be observed by the UWB beacon specific to Fig. 8(a). The IMU device is ICM-20649 and single-beacon range sensor uses LinkTrack P-B UWB specific to Fig. 8(b). A real-time kinematic (RTK) receiver is used as a high-accuracy reference for UAV specific to Fig. 8(c).

After an alignment period, the GPS/INS is started in a loosely coupled integrated navigation mode. One-round experiment takes approximately 1 minute to complete, throughout the experiment, the UWB beacon provides range measurements at a frequency of 20 Hz.

We compare the localization results obtained by fusing the GPS/INS navigation and UWB measurement through Algorithm 1 and EKF method in Fig. 9. Experimental results show that the proposed method accurately encompasses the UAV

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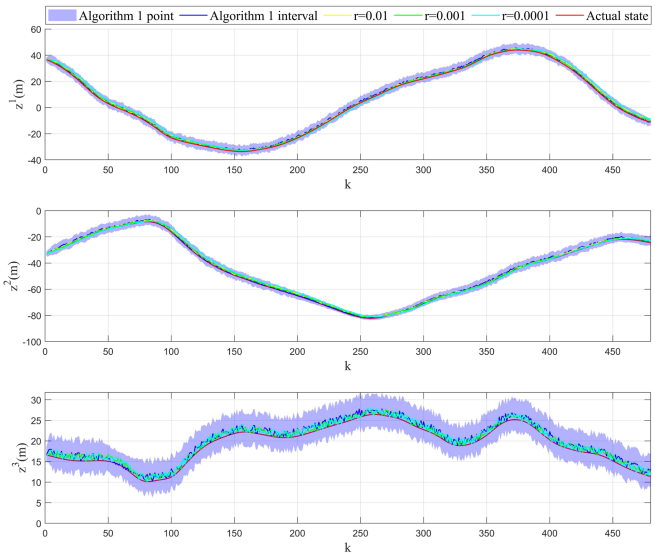



Fig. 9. Field flight experiment results. The horizontal coordinate represents the RTK sampling point with a sampling frequency of 10 Hz.

TABLE IV
COMPARISON FOR UAV LOCALIZATION

| Method | Algorithm 1 | r=0.01 | r=0.001 | r=0.0001 |
|---|---|---|---|---|
| MAE (m) | 1.78 | 1.97 | 1.59 | 2.84 |

display location. compared to EKF, Algorithm 1 does not need to know the exact distribution of the noise, and the quantitative results are provided in Table IV.

## V. CONCLUSION

In this letter, we have investigated an accurate and efficient set-membership filtering method based constrained zonotopes for the single-beacon localization problem. To handle the challenge of requiring accurate statistics of noises for previous filtering-based studies, we have proposed a new solution for single-beacon localization by set-membership filtering with UBB noises. We have designed a convex optimization to handle the non-convexity caused by the single-beacon range measurement and a halfspace-intersection refinement to improve the accuracy of localization. At the same time, a sliding-window recursive method has been proposed to ensure accuracy and efficiency. Simulation and field experiment have proved the effectiveness and the advantages of the SMF-based localization method. For future work, we will explore effective localization methods for specific classes of nonlinear models (e.g., the unicycle model).

## APPENDIX A
## PROOF OF THEOREM 1

From the definition of constrained zonotope, we obtain a representation formula for points in $\bar{\mathcal{X}}_k$. With (6), we have

$$\rho_k^i = \bar{g}_k^i \xi + \bar{c}_k^i. \tag{23}$$

By $[\![\mathbf{x}_k | y_{0:k-1}]\!] \subseteq \bar{\mathcal{X}}_k$, we get

$$\tilde{\mathcal{X}}_k = \{x_k \in \bar{\mathcal{X}}_k : y_k - \|z_k - p_k\| \in [\![\mathbf{v}_k]\!], z_k = Hx_k\}. \tag{24}$$

Denote by $\rho_k^i$ in (23) each component of $z_k$ in (24). Noting $y_k^- = y_k - \inf_{v_k \in [\![\mathbf{v}_k]\!]} v_k$ and $y_k^+ = y_k - \sup_{v_k \in [\![\mathbf{v}_k]\!]} v_k$, we transform $\|z_k - p_k\| = \sum_{i=1}^{n_z} \|\rho_k^i - p_k^i\| \subseteq [y_k^-, y_k^+]$ in (24) into the following form

$$\begin{aligned}\left(\sum_{i=1}^{n_z} \|\rho_k^i - p_k^i\|\right)^2 &= \xi^T \sum_{i=1}^{n_z} \left(\left(\bar{g}_k^i\right)^T \bar{g}_k^i\right) \xi \\ &\quad + 2 \sum_{i=1}^{n_z} \left(\bar{c}_k^i - p_k^i\right) \bar{g}_k^i \xi, \\ &\quad + \sum_{i=1}^{n_z} \left(\bar{c}_k^i - p_k^i\right)^2 \subseteq \left[(y^-)_k^2, (y^+)_k^2\right].\end{aligned} \tag{25}$$

When $(\sum_{i=1}^{n_z} \|\rho_k^i - p_k^i\|)^2 \geqslant (y_k^-)^2$, it is equivalent to

$$\begin{aligned}&\xi^T \sum_{i=1}^{n_z} ((\bar{g}_k^i)^T \bar{g}_k^i) \xi + 2 \sum_{i=1}^{n_z} (\bar{c}_k^i - p_k^i) \bar{g}_k^i \xi \\ &\quad + \sum_{i=1}^{n_z} (\bar{c}_k^i - p_k^i)^2 \geqslant (y_k^-)^2.\end{aligned} \tag{26}$$

By defining $Q_k^- = -\sum_{i=1}^{n_z} ((\bar{g}_k^i)^T \bar{g}_k^i)$, $q_k^- = -2 \sum_{i=1}^{n_z} (\bar{c}_k^i - p_k^i) \bar{g}_k^i$, $d_k^- = (y^-)_k^2 - \sum_{i=1}^{n_z} (\bar{c}_k^i - p_k^i)^2$, the formula (26) becomes

$$\xi^T Q_k^- \xi + q_k^- \xi + d_k^- \leqslant 0. \tag{27}$$

With a similar definition $Q_k^+ = \sum_{i=1}^{n_z} ((\bar{g}_k^i)^T \bar{g}_k^i)$, $q_k^+ = 2 \sum_{i=1}^{n_z} (\bar{c}_k^i - p_k^i) \bar{g}_k^i$, $d_k^+ = \sum_{i=1}^{n_z} (\bar{c}_k^i - p_k^i)^2 - (y_k^+)^2$, we can obtain:

$$\left(\sum_{i=1}^{n_z} \|\rho_k^i - p_k^i\|\right)^2 \leqslant (y_k^+)^2 \Longleftrightarrow \xi^T Q_k^+ \xi + q_k^+ \xi + d_k^+ \leqslant 0. \tag{28}$$

Clearly, $Q_k^-$ is a semi-negative definite matrix by its leading principal minors. Thus the range of constraints formed by (27) is non-convex. Finally, we prove Theorem 1.

## APPENDIX B
## PROOF OF THEOREM 2

Because $(Q_k^-)^T = Q_k^-$ is a real symmetric matrix, we have the eigenvalue decomposition as $Q_k^- = \sum_{\lambda_i \geqslant 0} \lambda_i \nu_i \nu_i^T + \sum_{\lambda_i < 0} \lambda_i \nu_i \nu_i^T = Q_k^* + \sum_{\lambda_i < 0} \lambda_i \nu_i \nu_i^T$. $Q_k^*$ is the semi-positive definite part of $Q_k^-$. As $\xi \in [\bar{h}_k^{x-}, \bar{h}_k^{x+}]$, we get $a_i^- \leqslant \nu_i^T \xi \leqslant a_i^+$. Then $(\nu_i^T \xi)^2 \leqslant (a_i^- + a_i^+) \nu_i^T \xi - a_i^- a_i^+$. From (27), we have

$$\xi^T \left(Q_k^* + \sum_{\lambda_i < 0} \lambda_i \nu_i \nu_i^T\right) \xi + q_k^- \xi + d_k^- \leqslant 0, \tag{29}$$

$$\Longleftrightarrow \xi^T Q_k^* \xi + q_k^- \xi + d_k^- \leqslant \xi^T \left(-\sum_{\lambda_i < 0} \lambda_i \nu_i \nu_i^T\right) \xi \tag{30}$$

$$\Longleftrightarrow \xi^T Q_k^* \xi + q_k^* \xi + d_k^* \leqslant 0, \tag{31}$$

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where $q_k^* = \sum_{\lambda_i<0} \lambda_i (a_i^- + a_i^+)\nu_i^T + q_k^-$ and $d_k^* = d_k^- - \sum_{\lambda_i<0} \lambda_i a_i^- a_i^+$. Clearly, it can be judged that $Q_k^+$ is a positive semi-definiteness matrix by its leading principal minors.

## Appendix C
## Proof of Theorem 3

Denote $\hat{\mathcal{X}}_k$ in the following intersection form

$$\hat{\mathcal{X}}_k = Z\left(\hat{G}_k, \hat{c}_k, \hat{A}_k, \hat{b}_k, \hat{h}_k\right) \tag{32}$$

$$= Z\left(\bar{G}_k^x, \bar{c}_k^x, \bar{A}_k^x, \bar{b}_k^x, \bar{h}_k^x\right) \bigcap\nolimits_H \left(\overline{\text{HS}}(L_k, F_k) \bigcap Z\left(\tilde{R}_k, \tilde{S}_k\right)\right). \tag{33}$$

myr The following form is obtained by applying Theorem 1 in [28] to each halfspace constrain in $\overline{\text{HS}}(L_k, F_k)$.

$$\overline{\text{HS}}(L_k, F_k) \bigcap Z\left(\tilde{R}_k, \tilde{S}_k\right) = Z\left(\left[\tilde{R}_k \ \mathbf{0}\right], \tilde{S}_k, \left[L_k \tilde{R}_k \ D_k\right],\right.$$

$$\left. F_k - L_k \tilde{S}_k - \frac{1}{2}\begin{bmatrix} d_k^1 & \dots & d_k^{n_p} \end{bmatrix}^T, I_{(n_z+n_p)\times 1}\right), \tag{34}$$

Finally, (18) is obtained with Proposition 1 in [25].

## Appendix D
## Proof of Theorem 4

First, with $k \in \mathbb{N}_0$, $\delta \le k$, we compute the interval hull of $\hat{\mathcal{X}}_{k-\delta}$ by linear programming in [25] as the following form

$$\hat{\mathcal{X}}_{k-\delta} = Z\left(\hat{G}_{k-\delta}^x, \hat{c}_{k-\delta}^x, \hat{A}_{k-\delta}^x, \hat{b}_{k-\delta}^x, \hat{h}_{k-\delta}^x\right) \subseteq \overline{\text{IH}}\left(\hat{\mathcal{X}}_{k-\delta}\right). \tag{35}$$

Second, based on (20) and the inclusion relation provided by (35) the following relation is obtained

$$\hat{\mathcal{X}}_k = \mathcal{F}_{k-\delta:k}\left(\hat{\mathcal{X}}_{k-\delta}\right) \subseteq \mathcal{F}_{k-\delta:k}\left(\overline{\text{IH}}\left(\hat{\mathcal{X}}_{k-\delta}\right)\right). \tag{36}$$

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